irAE-Map

Codebook and mapping rationale, version 1.0.0

A clinician-curated crosswalk of immune-related adverse events of immune checkpoint inhibitor therapy to CTCAE v6.0, ICD-10-CM FY2027, and FDA Adverse Event Reporting System reaction terms

Onyekachi Innocent Ike-Okpe

Independent Researcher

Built 2026-10-13. Status: released. DOI: https://doi.org/10.5281/zenodo.23351504. License: CC BY 4.0.

Contents

1. Summary
2. Sources
3. Methods
4. Results
5. Category rationale
6. Drug-cause companion codes
7. Limitations
8. Use of AI assistance
9. Versioning and maintenance
10. How to cite

1. Summary

irAE-Map links 36 immune-related adverse event (irAE) categories to the terms and codes that describe them in three vocabularies: the National Cancer Institute's Common Terminology Criteria for Adverse Events (CTCAE) version 6.0, the US diagnosis code set ICD-10-CM (fiscal year 2027), and the MedDRA reaction terms used in the FDA Adverse Event Reporting System (FAERS). Version 1.0.0 contains 918 mappings: 137 to CTCAE terms, 384 to ICD-10-CM code families covering 1,939 billable codes, and 397 to FAERS reaction terms. Each mapping has a role, a specificity rating, the author's confidence and a clinical rationale; Section 3.3 describes how the author decided them.

CTCAE v6.0, released by NCI in 2026, added terms in part to better describe adverse events of immunotherapy. Of the 134 CTCAE terms in irAE-Map, 29 are new in v6.0. ICD-10-CM gained a dedicated drug-cause code for immune checkpoint inhibitors, T45.AX5-, effective 1 October 2024. Health systems, registries and drug-safety analysts need a public, documented link between these vocabularies; irAE-Map provides one.

2. Sources

Source	Edition	Use in irAE-Map
NCI CTCAE spreadsheet (dctd.cancer.gov)	v6.0, MedDRA 28.0, errata 2026-01-20	Terms, MedDRA LLT codes, grade 1 to 5 definitions, v5 to v6 change flags
CDC/NCHS ICD-10-CM code descriptions and addenda (ftp.cdc.gov)	FY2027, effective 2026-10-01	Codes, billable flags, descriptions, codes new in FY2027
openFDA drug adverse event API (api.fda.gov)	FAERS data updated 2026-07-30; accessed 2026-10-13	Reaction terms reported with the 13 checkpoint inhibitors and report counts

Exact URLs, download dates, file sizes and SHA-256 checksums are in data/raw/PROVENANCE.txt in the repository.

3. Methods

3.1 Category frame

Categories follow the organ groupings of the ASCO 2021 guideline on management of irAEs in patients treated with immune checkpoint inhibitor therapy, with additions where CTCAE v6.0 or the literature identifies a distinct entity. Each category lists the diagnoses most often confused with it (Section 5).

3.2 Candidate generation

- CTCAE v6.0: terms named in each category's rules, matched exactly to the v6.0 spreadsheet; the build fails if a named term is absent.
- ICD-10-CM: codes under named code prefixes, plus codes whose description matches the category's keyword pattern, excluding keyword matches in the neoplasm, pregnancy, perinatal, injury, external-cause and Z chapters. Billable codes are grouped into code families (a named parent code with its children, or the remaining codes in a three-character block) for review, and expanded back to every billable code for release.
- FAERS: for each category keyword, openFDA count queries over reports naming any of the 13 checkpoint inhibitors (290,091 reports) and over all reports (20,692,687), keeping reaction terms that match the category pattern. Reports were matched by openFDA's

harmonized generic name or by the raw product name (generic or brand), because the harmonized field returned no reports for three of the drugs. Each kept term's counts are exact.

- Each candidate received a suggested role, confidence and reason from documented rules (repository file code/categories.py), drafted with AI assistance (Section 8). Rows the rules could not classify were left without a suggestion.

3.3 Author review

Decisions were recorded in a review workbook, one row per candidate. For version 1.0.0 the author accepted the drafted role, specificity and rationale as a group for 917 released mappings and set 1 individually (author's instruction of 13 October 2026; data/curation/DECISION_LOG.md, which also lists corrections made to drafts after spot checks). The 164 candidates without a drafted suggestion were not released. Each row records its decision type (column decision_type), reviewer and date. Individual review of accepted rows is planned for the next version.

Role	Meaning
primary	Names the condition itself; enough on its own to identify the category
supporting	A laboratory, symptom, severity or grading term consistent with, but not specific to, the category
lookalike	A different condition often confused with this one; listed so analysts exclude it deliberately
companion	ICD-10-CM drug-cause code that should accompany a condition code

Specificity	Meaning
High	Names the condition and is used for it in practice
Moderate	Right organ and process, but also used for non-immune causes
Low	Supporting or non-specific

Confidence is the author's confidence that a row's role is correct (including that a look-alike is a look-alike). The author rated every released row High. Specificity, not confidence, is the column to filter on when selecting codes.

3.4 Quality checks

The build verifies that every CTCAE term exists in v6.0 and every released ICD-10-CM code is a valid billable FY2027 code; that every released row has a role, a specificity rating (except look-alikes) and a rationale; that no suggestion-less row was accepted without a decision; and it reports, for each category, any vocabulary lacking a primary mapping. Results are in data/processed/qa_report.txt.

3.5 FAERS over-representation

For each reaction term, faers_ici_prr = (a / N_ici) / ((b - a) / (N_all - N_ici)), where a is the number of reports naming a checkpoint inhibitor with the term, b the number of all reports with the term, N_ici = 290,091 and N_all = 20,692,687. It describes over-representation only; it is not a disproportionality signal (no deduplication, stratification or confidence interval).

4. Results

Vocabulary	Mappings	Primary	Supporting	Look-alike	Companion
CTCAE v6.0	137	59	64	14	0
ICD-10-CM FY2027	384	142	32	208	2
FAERS MedDRA PT	397	224	105	68	0

Specificity: High 487, Moderate 294, Low 136. 8 terms are primary in more than one category; these are overlap entities such as thyroiditis (both thyroid categories) and the myocarditis-myositis-myasthenia gravis overlap syndrome.

5. Category rationale

For each category: the primary CTCAE terms with their grade 3 definition, primary ICD-10-CM code families, the most-reported primary FAERS terms, and look-alikes. The complete list, including supporting terms, is the released table.

Colitis / enterocolitis / diarrhea (COLITIS)

Organ system: Gastrointestinal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Colitis	Severe abdominal pain; peritoneal signs	High	CTCAE v6.0 term that names colitis / enterocolitis / diarrhea and grades its severity. CTCAE term for colon inflammation; the graded irAE term.
Enterocolitis	Severe or persistent abdominal pain; fever; ileus; peritoneal signs	High	CTCAE v6.0 term that names colitis / enterocolitis / diarrhea and grades its severity. Small and large bowel inflammation; ICI enterocolitis.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
K52.83*	Microscopic colitis	4	Moderate	ICD-10-CM code family that names colitis / enterocolitis / diarrhea. Microscopic colitis; reported phenotype of ICI colitis.
K52.9	Noninfective gastroenteritis and colitis, unspecified	1	Moderate	ICD-10-CM code family that names colitis / enterocolitis / diarrhea. Noninfective colitis, unspecified; common but non-specific.
K52.89	Other specified noninfective gastroenteritis and colitis	1	Moderate	ICD-10-CM code family that names colitis / enterocolitis / diarrhea. Other specified noninfective colitis; used for immune-mediated colitis.
K52.1	Toxic gastroenteritis and colitis	1	High	ICD-10-CM code family that names colitis / enterocolitis / diarrhea. Toxic gastroenteritis and colitis; ICD-10-CM index routes drug-induced colitis here.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Colitis	6,177	37,176	14	High
Immune-mediated enterocolitis	3,100	3,480	573.8	High
Enterocolitis	1,132	5,635	17.7	High
Autoimmune colitis	547	635	437.2	High
Colitis microscopic	207	4,632	3.3	Moderate

Look-alikes: Infectious colitis (C. difficile, CMV), pre-existing IBD flare, ischemic colitis, other drug diarrhea (e.g., concurrent TKI or chemotherapy).

Hepatitis (HEPATITIS)

Organ system: Gastrointestinal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Hepatitis (new in v6.0)	Symptomatic liver dysfunction; compensated cirrhosis; hospitalization or prolongation of existing hospitalization indicated	High	CTCAE v6.0 term that names hepatitis and grades its severity (new in v6.0). New in CTCAE v6.0; graded hepatitis term.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
K75.4	Autoimmune hepatitis	1	Moderate	ICD-10-CM code family that names hepatitis. Autoimmune hepatitis; also used for ICI hepatitis but describes a distinct disease.
K75.9	Inflammatory liver disease, unspecified	1	Low	ICD-10-CM code family that names hepatitis.
K71.2	Toxic liver disease with acute hepatitis	1	High	ICD-10-CM code family that names hepatitis.
K71.1*	Toxic liver disease with hepatic necrosis	2	Moderate	ICD-10-CM code family that names hepatitis.
K71.6	Toxic liver disease with hepatitis, not elsewhere classified	1	High	ICD-10-CM code family that names hepatitis. Toxic liver disease with hepatitis NEC.
K71.9	Toxic liver disease, unspecified	1	Moderate	ICD-10-CM code family that names hepatitis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Hepatitis	2,681	25,255	8.4	High
Immune-mediated hepatic disorder	1,830	1,887	2258	High
Immune-mediated hepatitis	1,814	1,938	1028.9	High
Drug-induced liver injury	1,357	30,439	3.3	High
Hepatic cytolysis	1,331	12,060	8.7	Moderate
Autoimmune hepatitis	1,187	6,833	14.8	High

Look-alikes: Viral hepatitis or HBV reactivation, hepatic metastases or progression, other hepatotoxic drugs (acetaminophen, azoles, TKIs), alcohol, biliary obstruction.

Pancreatitis / asymptomatic lipase elevation (PANCREATITIS)

Organ system: Gastrointestinal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Pancreatitis	Severe pain; vomiting; medical intervention indicated (e.g., analgesia, nutritional support)	High	CTCAE v6.0 term that names pancreatitis / asymptomatic lipase elevation and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
K85.9*	Acute pancreatitis, unspecified	3	Moderate	ICD-10-CM code family that names pancreatitis / asymptomatic lipase elevation.
K85.3*	Drug induced acute pancreatitis	3	High	ICD-10-CM code family that names pancreatitis / asymptomatic lipase elevation. Drug-induced acute pancreatitis.
K85.8*	Other acute pancreatitis	3	Moderate	ICD-10-CM code family that names pancreatitis / asymptomatic lipase elevation. Other acute pancreatitis (includes autoimmune acute pancreatitis).

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Pancreatitis	1,293	52,220	1.8	High
Pancreatitis acute	410	22,020	1.3	High
Immune-mediated pancreatitis	355	364	2774.2	High
Autoimmune pancreatitis	151	498	30.6	High
Oedematous pancreatitis	30	551	4	Moderate

Look-alikes: Gallstone or alcohol pancreatitis, hypertriglyceridemia, pancreatic metastases; isolated enzyme rise without symptoms is common and often not treated.

Pneumonitis (PNEUMONITIS)

Organ system: Pulmonary

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Pneumonitis	Severe symptoms; oxygen indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names pneumonitis and grades its severity. Graded term; most common fatal ICI toxicity in some series.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
J70.2	Acute drug-induced interstitial lung disorders	1	High	ICD-10-CM code family that names pneumonitis.
J70.3	Chronic drug-induced interstitial lung disorders	1	High	ICD-10-CM code family that names pneumonitis.
J84.116	Cryptogenic organizing pneumonia	1	Moderate	ICD-10-CM code family that names pneumonitis. Cryptogenic organizing pneumonia; organizing pneumonia is an ICI pattern.
J70.4	Drug-induced interstitial lung disorders, unspecified	1	High	ICD-10-CM code family that names pneumonitis.
J84.9	Interstitial pulmonary disease, unspecified	1	Moderate	ICD-10-CM code family that names pneumonitis.
J84.89	Other specified interstitial pulmonary diseases	1	Moderate	ICD-10-CM code family that names pneumonitis. Other specified ILD.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Pneumonitis	6,132	26,681	21	High
Interstitial lung disease	5,821	47,680	9.8	High
Immune-mediated lung disease	1,618	1,725	1063.5	High
Organising pneumonia	439	4,843	7	High
Immune-mediated pneumonitis	220	234	1105.2	High
Autoimmune lung disease	80	114	165.5	High

Look-alikes: Infectious pneumonia (including PJP), radiation pneumonitis, lymphangitic tumor spread, pulmonary edema, pulmonary embolism, COPD exacerbation.

Thyroiditis / primary hypothyroidism (HYPOTHYROIDISM)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Hypothyroidism	Severe symptoms; hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names thyroiditis / primary hypothyroidism and grades its severity.
Thyroiditis (new in v6.0)	Hospitalization indicated	High	CTCAE v6.0 term that names thyroiditis / primary hypothyroidism and grades its severity (new in v6.0). New in v6.0; captures the destructive thyroiditis that precedes hypothyroidism.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E06.3	Autoimmune thyroiditis	1	Moderate	ICD-10-CM code family that names thyroiditis / primary hypothyroidism. Autoimmune thyroiditis; also pre-

ICD-10-CM	Description	Billable codes	Specificity	Rationale
				existing Hashimoto.
E06.4	Drug-induced thyroiditis	1	High	ICD-10-CM code family that names thyroiditis / primary hypothyroidism.
E03.2	Hypothyroidism due to medicaments and other exogenous substances	1	High	ICD-10-CM code family that names thyroiditis / primary hypothyroidism.
E03.9	Hypothyroidism, unspecified	1	Moderate	ICD-10-CM code family that names thyroiditis / primary hypothyroidism. Hypothyroidism, unspecified; very common at baseline.
E06.5	Other chronic thyroiditis	1	Low	ICD-10-CM code family that names thyroiditis / primary hypothyroidism.
E03.8	Other specified hypothyroidism	1	Moderate	ICD-10-CM code family that names thyroiditis / primary hypothyroidism.
E06.9	Thyroiditis, unspecified	1	Moderate	ICD-10-CM code family that names thyroiditis / primary hypothyroidism.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Thyroiditis	1,281	2,941	54.3	High
Immune-mediated hypothyroidism	1,046	1,061	4904.5	High
Immune-mediated thyroiditis	458	463	6442.4	High
Autoimmune thyroiditis	394	3,715	8.3	High
Hypothyroidism	199	1,971	7.9	High
Autoimmune hypothyroidism	138	231	104.4	High

Look-alikes: Pre-existing hypothyroidism, post-surgical or post-radiation hypothyroidism, sick euthyroid syndrome, central hypothyroidism from hypophysitis (low TSH).

Thyroiditis / hyperthyroidism (HYPERTHYROIDISM)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Hyperthyroidism	Severe symptoms; hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names thyroiditis / hyperthyroidism and grades its severity. Graded term; usually transient thyrotoxic phase of thyroiditis.
Thyroiditis (new in v6.0)	Hospitalization indicated	High	CTCAE v6.0 term that names thyroiditis / hyperthyroidism and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E06.2	Chronic thyroiditis with transient thyrotoxicosis	1	Moderate	ICD-10-CM code family that names thyroiditis / hyperthyroidism.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E06.4	Drug-induced thyroiditis	1	High	ICD-10-CM code family that names thyroiditis / hyperthyroidism.
E05.8*	Other thyrotoxicosis	2	Moderate	ICD-10-CM code family that names thyroiditis / hyperthyroidism.
E06.1	Subacute thyroiditis	1	Low	ICD-10-CM code family that names thyroiditis / hyperthyroidism. Subacute thyroiditis; painful viral thyroiditis is a look-alike.
E05.9*	Thyrotoxicosis, unspecified	2	Moderate	ICD-10-CM code family that names thyroiditis / hyperthyroidism.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Hyperthyroidism	2,545	14,275	15.3	High
Thyroiditis	1,281	2,941	54.3	High
Immune-mediated thyroiditis	458	463	6442.4	High
Autoimmune thyroiditis	394	3,715	8.3	High
Immune-mediated hyperthyroidism	231	236	3249.3	High
Silent thyroiditis	100	142	167.5	High

Look-alikes: Iodinated contrast-induced thyrotoxicosis, toxic nodules, exogenous levothyroxine excess, viral subacute thyroiditis.

Hypophysitis / hypopituitarism (HYPOPHYSITIS)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Hypophysitis	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names hypophysitis / hypopituitarism and grades its severity. Graded term; typical of anti-CTLA-4.
Hypopituitarism	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names hypophysitis / hypopituitarism and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E23.7	Disorder of pituitary gland, unspecified	1	Low	ICD-10-CM code family that names hypophysitis / hypopituitarism.
E23.1	Drug-induced hypopituitarism	1	High	ICD-10-CM code family that names hypophysitis / hypopituitarism.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E23.0	Hypopituitarism	1	Moderate	ICD-10-CM code family that names hypophysitis / hypopituitarism. Hypopituitarism (other).
E23.6	Other disorders of pituitary gland	1	High	ICD-10-CM code family that names hypophysitis / hypopituitarism. Other disorders of pituitary gland; ICD-10-CM index routes hypophysitis here.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Hypophysitis	1,855	1,986	995.9	High
Hypopituitarism	929	1,720	82.6	High
Adrenocorticotropic hormone deficiency	760	861	529.2	High
Hypothalamo-pituitary disorder	441	1,425	31.5	Moderate
Immune-mediated hypophysitis	416	423	4179.7	High
Lymphocytic hypophysitis	219	273	285.2	High

Look-alikes: Pituitary metastasis or adenoma, exogenous glucocorticoid suppression of the HPA axis, primary adrenal insufficiency.

Primary adrenal insufficiency (adrenalitis) (ADRENAL_INSUFFICIENCY)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Adrenal insufficiency	Severe symptoms; hospitalization indicated	High	CTCAE v6.0 term that names primary adrenal insufficiency and grades its severity. Graded term; covers primary and secondary, so category needs context.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E27.3	Drug-induced adrenocortical insufficiency	1	Moderate	ICD-10-CM code family that names primary adrenal insufficiency. Drug-induced adrenocortical insufficiency; also used for steroid withdrawal.
E27.49	Other adrenocortical insufficiency	1	Low	ICD-10-CM code family that names primary adrenal insufficiency.
E27.1	Primary adrenocortical insufficiency	1	High	Primary adrenocortical insufficiency; ICI adrenalitis is a primary adrenal insufficiency, so this code names the condition
E27.40	Unspecified adrenocortical insufficiency	1	Low	ICD-10-CM code family that names primary adrenal insufficiency.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Adrenal insufficiency	3,896	11,815	34.6	High
Immune-mediated adrenal insufficiency	739	745	8662.5	High
Addison^s disease	144	987	12	Moderate
Primary adrenal insufficiency	94	294	33.1	High
Adrenalitis	75	105	175.8	High

Look-alikes: Secondary adrenal insufficiency from hypophysitis (see HYPOPHYSITIS), glucocorticoid withdrawal, adrenal metastases or hemorrhage.

Immune checkpoint inhibitor diabetes (insulin-deficient) (DIABETES)

Organ system: Endocrine

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Diabetes mellitus (new in v6.0)	Insulin therapy initiated or increased; hospitalization indicated	High	CTCAE v6.0 term that names immune checkpoint inhibitor diabetes and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
E09*	Drug or chemical induced diabetes mellitus	86	Moderate	ICD-10-CM code family that names immune checkpoint inhibitor diabetes. Drug-induced diabetes; but most often steroid-induced hyperglycemia.
E13*	Other specified diabetes mellitus	86	Moderate	ICD-10-CM code family that names immune checkpoint inhibitor diabetes. Other specified diabetes.
E10*	Type 1 diabetes mellitus	82	High	ICD-10-CM code family that names immune checkpoint inhibitor diabetes. Type 1 diabetes; ICI diabetes is insulin-deficient and usually coded here.
E10.1*	Type 1 diabetes mellitus with ketoacidosis	2	High	ICD-10-CM code family that names immune checkpoint inhibitor diabetes. Type 1 diabetes with ketoacidosis; common presentation.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Type 1 diabetes mellitus	1,601	5,929	26	High
Diabetes mellitus	1,555	75,562	1.5	High
Diabetic ketoacidosis	1,461	23,655	4.6	High
Fulminant type 1 diabetes mellitus	838	1,007	348.7	High

Look-alikes: Steroid-induced hyperglycemia (very common during irAE treatment), pre-existing type 2 diabetes, pancreatogenic diabetes.

Myocarditis (MYOCARDITIS)

Organ system: Cardiovascular

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Myocarditis	Severe with symptoms at rest or with minimal activity or exertion; intervention indicated; new onset of symptoms	High	CTCAE v6.0 term that names myocarditis and grades its severity. Graded term; high fatality irAE.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
I40.9	Acute myocarditis, unspecified	1	High	ICD-10-CM code family that names myocarditis.
I40.1	Isolated myocarditis	1	High	ICD-10-CM code family that names myocarditis.
I51.4	Myocarditis, unspecified	1	Moderate	ICD-10-CM code family that names myocarditis.
I40.8	Other acute myocarditis	1	High	ICD-10-CM code family that names myocarditis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Myocarditis	3,053	11,175	26.4	High
Immune-mediated myocarditis	1,441	1,519	1299.3	High
Myocarditis-myositis-myasthenia gravis overlap syndrome	174	179	2447.5	High
Autoimmune myocarditis	149	216	156.4	High

Look-alikes: Acute coronary syndrome, viral myocarditis, takotsubo, cardiotoxicity from anthracyclines or HER2 therapy; often overlaps with myositis and myasthenia (see MYOSITIS, MYASTHENIA).

Pericarditis / pericardial effusion (PERICARDITIS)

Organ system: Cardiovascular

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Pericarditis	Pericarditis with physiologic consequences (e.g., pericardial constriction)	High	CTCAE v6.0 term that names pericarditis / pericardial effusion and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
I30.0	Acute nonspecific idiopathic pericarditis	1	Moderate	ICD-10-CM code family that names pericarditis / pericardial effusion.
I30.9	Acute pericarditis, unspecified	1	High	ICD-10-CM code family that names pericarditis / pericardial effusion.
I31.9	Disease of pericardium, unspecified	1	Low	ICD-10-CM code family that names pericarditis / pericardial effusion.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
I30.8	Other forms of acute pericarditis	1	High	ICD-10-CM code family that names pericarditis / pericardial effusion.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Pericarditis	407	16,450	1.8	High
Immune-mediated pericarditis	52	54	1828.6	High
Autoimmune pericarditis	12	20	105.5	High

Look-alikes: Malignant pericardial effusion (very common in lung cancer), radiation pericarditis, infection, uremia.

Vasculitis (VASCULITIS)

Organ system: Cardiovascular

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Vasculitis	Severe symptoms, medical intervention indicated (e.g., steroids)	High	CTCAE v6.0 term that names vasculitis and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
I77.6	Arteritis, unspecified	1	Moderate	ICD-10-CM code family that names vasculitis.
M31.5	Giant cell arteritis with polymyalgia rheumatica	1	Moderate	ICD-10-CM code family that names vasculitis.
M31.9	Necrotizing vasculopathy, unspecified	1	Moderate	ICD-10-CM code family that names vasculitis.
M31.6	Other giant cell arteritis	1	Moderate	ICD-10-CM code family that names vasculitis.
M31.8	Other specified necrotizing vasculopathies	1	Moderate	ICD-10-CM code family that names vasculitis.
L95*	Vasculitis limited to skin, not elsewhere classified	4	Moderate	ICD-10-CM code family that names vasculitis. Vasculitis limited to skin.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Vasculitis	260	11,511	1.6	Moderate
Cutaneous vasculitis	88	3,524	1.8	Moderate
Giant cell arteritis	49	797	4.6	Moderate
Central nervous system vasculitis	40	482	6.4	Moderate
Temporal arteritis	34	927	2.7	Moderate
Hypersensitivity vasculitis	29	2,622	0.8	Moderate

Look-alikes: Infection, drug hypersensitivity vasculitis from other drugs, paraneoplastic vasculitis.

Nephritis (acute interstitial / glomerulonephritis) (NEPHRITIS)

Organ system: Renal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Glomerulonephritis (new in v6.0)	IV therapy indicated; immunosuppressant indicated; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names nephritis and grades its severity (new in v6.0). New in v6.0; less common with ICIs.
Tubulointerstitial nephritis (new in v6.0)	IV therapy indicated; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names nephritis and grades its severity (new in v6.0). New in v6.0; the typical ICI renal lesion.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
N00*	Acute nephritic syndrome	13	Low	ICD-10-CM code family that names nephritis.
N14.19	Nephropathy induced by other drugs, medicaments and biological substances	1	High	ICD-10-CM code family that names nephritis. Nephropathy induced by other drugs (not contrast).
N12	Tubulo-interstitial nephritis, not specified as acute or chronic	1	Moderate	ICD-10-CM code family that names nephritis.
N05*	Unspecified nephritic syndrome	11	Low	ICD-10-CM code family that names nephritis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Tubulointerstitial nephritis	1,834	18,485	7.7	High
Nephritis	869	3,266	25.5	High
Nephropathy toxic	430	10,992	2.9	Moderate
Immune-mediated nephritis	412	447	827.9	High
Nephrotic syndrome	403	6,761	4.5	Moderate
Immune-mediated renal disorder	372	386	1868.8	High

Look-alikes: Prerenal AKI (dehydration from colitis), contrast nephropathy, other nephrotoxins (PPIs, NSAIDs, platinum), obstruction, pyelonephritis.

Inflammatory dermatitis / rash / pruritus (DERM_INFLAMMATORY)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Dermatitis psoriasiform (new in v6.0)	IV intervention indicated	High	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity (new in v6.0). New in v6.0;

CTCAE v6.0 term	Grade 3	Specificity	Rationale
			characteristic ICI pattern.
Eczematous dermatitis (new in v6.0)	IV intervention indicated	High	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity (new in v6.0).
Erythroderma	Erythema covering >90% BSA with associated symptoms (e.g., pruritus or tenderness); limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity. Severe diffuse dermatitis.
Exfoliative dermatitis (new in v6.0)	IV intervention indicated	Moderate	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity (new in v6.0).
Lichenoid dermatitis (new in v6.0)	IV intervention indicated	High	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity (new in v6.0). New in v6.0; characteristic ICI pattern.
Papulovesicular rash (new in v6.0)	Covering >50% BSA; moderate or severe symptoms	Moderate	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity (new in v6.0).
Rash maculo-papular	Macules/papules covering >50% BSA; moderate or severe symptoms	High	CTCAE v6.0 term that names inflammatory dermatitis / rash / pruritus and grades its severity. Most common ICI skin irAE.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
L30.9	Dermatitis, unspecified	1	Low	ICD-10-CM code family that names inflammatory dermatitis / rash / pruritus.
L26	Exfoliative dermatitis	1	Moderate	ICD-10-CM code family that names inflammatory dermatitis / rash / pruritus.
L27.0	Generalized skin eruption due to drugs and medicaments taken internally	1	High	ICD-10-CM code family that names inflammatory dermatitis / rash / pruritus. Generalized skin eruption due to drugs taken internally.
L43.2	Lichenoid drug reaction	1	High	ICD-10-CM code family that names inflammatory dermatitis / rash / pruritus.
L27.1	Localized skin eruption due to drugs and medicaments taken internally	1	High	ICD-10-CM code family that names inflammatory dermatitis / rash / pruritus. Localized skin eruption due to drugs taken internally.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Rash	9,000	427,888	1.5	High
Rash maculo-papular	1,402	21,868	4.8	High
Drug eruption	1,110	17,396	4.8	Moderate
Rash pruritic	1,014	53,237	1.4	High
Immune-mediated dermatitis	891	924	1899	High
Dermatitis	766	20,447	2.7	High

Look-alikes: EGFR inhibitor acneiform rash, radiation dermatitis, other drug eruptions (antibiotics), infection (scabies, fungal), xerosis.

Bullous dermatoses (bullous pemphigoid) (DERM_BULLOUS)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Dermatosis bullous (new in v6.0)	Blisters covering >50% BSA; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names bullous dermatoses and grades its severity (new in v6.0). New in v6.0; bullous pemphigoid is the typical ICI entity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
L13.9	Bullous disorder, unspecified	1	Low	ICD-10-CM code family that names bullous dermatoses.
L12.0	Bullous pemphigoid	1	High	ICD-10-CM code family that names bullous dermatoses.
L12.8	Other pemphigoid	1	Moderate	ICD-10-CM code family that names bullous dermatoses.
L12.9	Pemphigoid, unspecified	1	Moderate	ICD-10-CM code family that names bullous dermatoses.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Pemphigoid	1,279	7,314	14.9	High
Dermatitis bullous	419	7,100	4.4	High
Mucous membrane pemphigoid	19	103	15.9	High

Look-alikes: Bullous drug eruptions from other drugs, SJS/TEN (see DERM_SCAR), pemphigus, friction or edema blisters.

Severe cutaneous adverse reactions (SJS/TEN, DRESS, AGEP) (DERM_SCAR)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Erythema multiforme	Associated with oral or genital erosions	Moderate	CTCAE v6.0 term that names severe cutaneous adverse

CTCAE v6.0 term	Grade 3	Specificity	Rationale
			reactions and grades its severity.
Pustular drug eruption (new in v6.0)	Severe symptoms; IV antibiotic or steroid indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names severe cutaneous adverse reactions and grades its severity (new in v6.0). New in v6.0; AGEP.
Stevens-Johnson syndrome	Skin sloughing covering <10% BSA	High	CTCAE v6.0 term that names severe cutaneous adverse reactions and grades its severity.
Toxic epidermal necrolysis	-	High	CTCAE v6.0 term that names severe cutaneous adverse reactions and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D72.12	Drug rash with eosinophilia and systemic symptoms syndrome	1	High	ICD-10-CM code family that names severe cutaneous adverse reactions. Drug rash with eosinophilia and systemic symptoms (DRESS).
L51.9	Erythema multiforme, unspecified	1	Moderate	ICD-10-CM code family that names severe cutaneous adverse reactions.
L51.0	Nonbullous erythema multiforme	1	Moderate	ICD-10-CM code family that names severe cutaneous adverse reactions.
L51.8	Other erythema multiforme	1	Moderate	ICD-10-CM code family that names severe cutaneous adverse reactions.
L51.1	Stevens-Johnson syndrome	1	High	ICD-10-CM code family that names severe cutaneous adverse reactions.
L51.3	Stevens-Johnson syndrome-toxic epidermal necrolysis overlap syndrome	1	High	ICD-10-CM code family that names severe cutaneous adverse reactions. SJS-TEN overlap.
L51.2	Toxic epidermal necrolysis [Lyell]	1	High	ICD-10-CM code family that names severe cutaneous adverse reactions. Toxic epidermal necrolysis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Stevens-johnson syndrome	1,003	21,990	3.4	High
Toxic epidermal necrolysis	974	14,884	4.9	High
Erythema multiforme	687	9,202	5.7	High
Drug reaction with eosinophilia and systemic symptoms	324	25,217	0.9	High
Acute generalised exanthematous pustulosis	93	7,636	0.9	High

Look-alikes: SCAR from co-medications (allopurinol, sulfonamides, anticonvulsants, antibiotics) is often more likely than from the ICI.

Vitiligo / depigmentation (VITILIGO)

Organ system: Dermatologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Skin hypopigmentation	-	High	CTCAE v6.0 term that names vitiligo / depigmentation and grades its severity. CTCAE term covering vitiligo.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
L81.5	Leukoderma, not elsewhere classified	1	Moderate	ICD-10-CM code family that names vitiligo / depigmentation. Leukoderma, NEC.
L81.6	Other disorders of diminished melanin formation	1	Low	ICD-10-CM code family that names vitiligo / depigmentation.
L80	Vitiligo	1	High	ICD-10-CM code family that names vitiligo / depigmentation.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Vitiligo	508	2,275	20.2	High
Leukoderma	119	197	107.3	High
Skin depigmentation	43	1,341	2.3	High
Skin hypopigmentation	40	1,731	1.7	High
Retinal depigmentation	6	302	1.4	High

Look-alikes: Pre-existing vitiligo, post-inflammatory hypopigmentation.

Inflammatory arthritis (ARTHRITIS)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Arthritis	Severe pain associated with signs of inflammation, erythema, or joint swelling; irreversible joint damage; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names inflammatory arthritis and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
M13.8*	Other specified arthritis	23	Moderate	ICD-10-CM code family that names inflammatory arthritis.
M13.80	Other specified arthritis, unspecified site	1	Moderate	ICD-10-CM code family that names inflammatory arthritis.
M13.0	Polyarthritis, unspecified	1	Moderate	ICD-10-CM code family that names inflammatory arthritis.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
M06.00	Rheumatoid arthritis without rheumatoid factor, unspecified site	1	Moderate	ICD-10-CM code family that names inflammatory arthritis. Seronegative rheumatoid arthritis, unspecified site.
M06.9	Rheumatoid arthritis, unspecified	1	Moderate	ICD-10-CM code family that names inflammatory arthritis. Rheumatoid arthritis, unspecified; used for ICI inflammatory arthritis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Arthritis	1,215	77,756	1.1	High
Rheumatoid arthritis	890	111,584	0.6	High
Polyarthritis	430	5,184	6.4	High
Immune-mediated arthritis	350	365	1641.1	High
Autoimmune arthritis	82	153	81.2	High
Seronegative arthritis	65	1,108	4.4	High

Look-alikes: Osteoarthritis, gout or pseudogout, septic arthritis, pre-existing rheumatoid arthritis flare, bone metastases.

Polymyalgia rheumatica-like syndrome (PMR)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Musculoskeletal and connective tissue disorder - Other, specify	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names polymyalgia rheumatica-like syndrome and grades its severity. CTCAE has no PMR term.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
M31.5	Giant cell arteritis with polymyalgia rheumatica	1	Moderate	ICD-10-CM code family that names polymyalgia rheumatica-like syndrome. Giant cell arteritis with PMR.
M35.3	Polymyalgia rheumatica	1	High	ICD-10-CM code family that names polymyalgia rheumatica-like syndrome.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Polymyalgia rheumatica	363	3,074	9.4	High

Look-alikes: Myositis (CK elevated), hypothyroid myopathy, bone metastases, statin myalgia.

Sicca syndrome / Sjogren-like (SICCA)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Salivary duct inflammation	Acute salivary gland necrosis; severe secretion-induced symptoms (e.g., thick saliva/oral secretions or gagging); tube feeding or TPN indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names sicca syndrome / sjogren-like and grades its severity. Sialadenitis.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
K11.2*	Sialoadenitis	4	Moderate	ICD-10-CM code family that names sicca syndrome / sjogren-like.
M35.0*	Sjogren syndrome	13	High	ICD-10-CM code family that names sicca syndrome / sjogren-like. Sicca syndrome (Sjogren).

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Sjogren^s syndrome	336	4,414	5.8	High
Sialoadenitis	126	1,340	7.3	Moderate
Sicca syndrome	34	172	17.3	High
Noninfective sialoadenitis	4	157	1.8	Moderate

Look-alikes: Radiation xerostomia (head and neck RT), anticholinergic drugs, pre-existing Sjogren.

Myositis (MYOSITIS)

Organ system: Musculoskeletal

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Myositis	Pain associated with severe weakness; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names myositis and grades its severity. Graded term; overlaps with myocarditis and myasthenia.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
M60.9	Myositis, unspecified	1	High	ICD-10-CM code family that names myositis.
M33.1*	Other dermatomyositis	5	Moderate	ICD-10-CM code family that names myositis.
G72.49	Other inflammatory and immune myopathies, not elsewhere classified	1	Moderate	ICD-10-CM code family that names myositis. Other inflammatory and immune myopathies.
M60.8*	Other myositis	24	High	ICD-10-CM code family that names

ICD-10-CM	Description	Billable codes	Specificity	Rationale
				myositis.
M33.2*	Polymyositis	4	Moderate	ICD-10-CM code family that names myositis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Myositis	1,859	8,854	18.7	High
Immune-mediated myositis	732	2,603	27.5	High
Myopathy	292	8,551	2.5	Moderate
Myocarditis-myositis-myasthenia gravis overlap syndrome	174	179	2447.5	High
Polymyositis	126	1,744	5.5	High
Autoimmune myositis	86	452	16.5	High

Look-alikes: Statin myopathy, steroid myopathy, hypothyroid myopathy, paraneoplastic myositis; screen for concurrent myocarditis and myasthenia.

Myasthenia gravis / myasthenic syndrome (MYASTHENIA)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Myasthenia gravis	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names myasthenia gravis / myasthenic syndrome and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
G70.80	Lambert-Eaton syndrome, unspecified	1	Low	ICD-10-CM code family that names myasthenia gravis / myasthenic syndrome.
G70.01	Myasthenia gravis with (acute) exacerbation	1	High	ICD-10-CM code family that names myasthenia gravis / myasthenic syndrome. Myasthenia gravis with exacerbation.
G70.00	Myasthenia gravis without (acute) exacerbation	1	High	ICD-10-CM code family that names myasthenia gravis / myasthenic syndrome. Myasthenia gravis without exacerbation.
G70.1	Toxic myoneural disorders	1	Moderate	ICD-10-CM code family that names myasthenia gravis / myasthenic syndrome.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Myasthenia gravis	1,294	7,772	14	High
Immune-mediated myasthenia gravis	289	301	1693.8	High
Myasthenic syndrome	260	752	37.2	High
Myocarditis-myositis-myasthenia gravis overlap syndrome	174	179	2447.5	High
Myasthenia gravis crisis	78	1,415	4.1	High
Ocular myasthenia	48	340	11.6	High

Look-alikes: Pre-existing MG, Lambert-Eaton (paraneoplastic, small cell lung cancer), myositis with ocular involvement, brainstem lesions.

Guillain-Barre syndrome / inflammatory polyradiculoneuropathy (GBS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Guillain-Barre syndrome	Severe symptoms; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names guillain-barre syndrome / inflammatory polyradiculoneuropathy and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
G61.81	Chronic inflammatory demyelinating polyneuritis	1	Moderate	ICD-10-CM code family that names guillain-barre syndrome / inflammatory polyradiculoneuropathy.
G61.0	Guillain-Barre syndrome	1	High	ICD-10-CM code family that names guillain-barre syndrome / inflammatory polyradiculoneuropathy.
G61.9	Inflammatory polyneuropathy, unspecified	1	Low	ICD-10-CM code family that names guillain-barre syndrome / inflammatory polyradiculoneuropathy.
G61.89	Other inflammatory polyneuropathies	1	Moderate	ICD-10-CM code family that names guillain-barre syndrome / inflammatory polyradiculoneuropathy.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Guillain-barre syndrome	517	4,761	8.6	High
Demyelinating polyneuropathy	82	1,016	6.2	High
Chronic inflammatory demyelinating polyradiculoneuropathy	75	1,400	4	High
Miller fisher syndrome	43	223	16.8	High
Polyradiculoneuropathy	18	39	60.3	High
Axonal and demyelinating polyneuropathy	16	92	14.8	High

Look-alikes: Chemotherapy neuropathy, leptomeningeal disease, spinal cord compression, critical illness polyneuropathy.

Peripheral, cranial, and autonomic neuropathy (PERIPHERAL_NEUROPATHY)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Autonomic neuropathy (new in v6.0)	Severe symptoms; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names peripheral, cranial, and autonomic neuropathy and grades its severity (new in v6.0).
Facial nerve disorder	Severe symptoms; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names peripheral, cranial, and autonomic neuropathy and grades its severity. Facial palsy is a reported ICI neuropathy.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
G51.0	Bell's palsy	1	Moderate	ICD-10-CM code family that names peripheral, cranial, and autonomic neuropathy. Bell palsy.
G90.9	Disorder of the autonomic nervous system, unspecified	1	Low	ICD-10-CM code family that names peripheral, cranial, and autonomic neuropathy. Disorder of autonomic nervous system, unspecified.
G62.0	Drug-induced polyneuropathy	1	Moderate	ICD-10-CM code family that names peripheral, cranial, and autonomic neuropathy. Drug-induced polyneuropathy; usually chemotherapy (platinum, taxane).
G62.9	Polyneuropathy, unspecified	1	Low	ICD-10-CM code family that names peripheral, cranial, and autonomic neuropathy.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Facial paralysis	374	8,372	3.3	Moderate
Immune-mediated neuropathy	119	144	334.8	High

Look-alikes: Chemotherapy-induced neuropathy (very common in combination regimens), diabetic neuropathy, B12 deficiency, tumor compression.

Aseptic meningitis (ASEPTIC_MENINGITIS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Aseptic meningitis (new in v6.0)	Steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	CTCAE v6.0 term that names aseptic meningitis and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
G03.8	Meningitis due to other specified causes	1	Moderate	ICD-10-CM code family that names aseptic meningitis.
G03.9	Meningitis, unspecified	1	Low	ICD-10-CM code family that names aseptic meningitis.
G03.0	Nonpyogenic meningitis	1	High	ICD-10-CM code family that names aseptic meningitis. Nonpyogenic meningitis (aseptic).

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Meningitis	387	6,596	4.4	Moderate
Meningitis aseptic	377	4,416	6.6	High
Meningitis noninfective	9	62	11.9	High

Look-alikes: Infectious meningitis, leptomeningeal carcinomatosis, drug-induced aseptic meningitis (NSAIDs, IVIG).

Encephalitis / meningoencephalitis (ENCEPHALITIS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Encephalitis (new in v6.0)	IV antibiotic, antifungal, or antiviral intervention indicated; steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	CTCAE v6.0 term that names encephalitis / meningoencephalitis and grades its severity (new in v6.0).
Encephalomyelitis (new in v6.0)	IV antibiotic, antifungal, or antiviral intervention indicated; steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	CTCAE v6.0 term that names encephalitis / meningoencephalitis and grades its severity (new in v6.0).
Meningoencephalitis (new in v6.0)	IV antibiotic, antifungal, or antiviral intervention indicated; steroid therapy indicated; severe changes in mental status; self-limited seizure activity; neurologic abnormalities	High	CTCAE v6.0 term that names encephalitis / meningoencephalitis and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
G04.00	Acute disseminated encephalitis and encephalomyelitis, unspecified	1	Moderate	ICD-10-CM code family that names encephalitis / meningoencephalitis. Acute disseminated encephalitis, unspecified.
G04.90	Encephalitis and encephalomyelitis, unspecified	1	Moderate	ICD-10-CM code family that names encephalitis / meningoencephalitis.
G04.81	Other encephalitis and encephalomyelitis	1	High	ICD-10-CM code family that names encephalitis / meningoencephalitis.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
				Other encephalitis and encephalomyelitis (includes autoimmune encephalitis).

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Encephalitis	1,094	6,807	13.5	High
Immune-mediated encephalitis	457	489	1004.4	High
Encephalitis autoimmune	353	1,276	26.9	High
Limbic encephalitis	81	276	29.2	High
Noninfective encephalitis	40	706	4.2	High

Look-alikes: HSV or other viral encephalitis, brain metastases, PRES, metabolic encephalopathy, ICANS after CAR-T.

Transverse myelitis (MYELITIS)

Organ system: Neurologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Myelitis	Severe weakness or sensory loss; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names transverse myelitis and grades its severity. Graded term (Infections SOC in CTCAE, but non-infectious myelitis also graded here).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
G37.3	Acute transverse myelitis in demyelinating disease of central nervous system	1	High	ICD-10-CM code family that names transverse myelitis. Acute transverse myelitis in demyelinating disease.
G04.91	Myelitis, unspecified	1	Moderate	ICD-10-CM code family that names transverse myelitis.
G04.89	Other myelitis	1	Moderate	ICD-10-CM code family that names transverse myelitis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Myelitis	100	1,533	4.9	High
Myelitis transverse	98	1,391	5.3	High
Immune-mediated myelitis	15	23	131.9	High

Look-alikes: Epidural metastasis or cord compression, radiation myelopathy, infection, multiple sclerosis or NMOSD.

Uveitis / scleritis / episcleritis (OCULAR_INFLAMMATION)

Organ system: Ocular

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Scleral disorder	Symptomatic with marked decrease in visual acuity (best corrected visual acuity worse than 20/40 or more than 3 lines of decreased vision from known baseline, up to 20/200); limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names uveitis / scleritis / episcleritis and grades its severity. Scleritis/episcleritis.
Uveitis	Anterior uveitis with 3+ or greater cells; intermediate posterior or pan-uveitis	High	CTCAE v6.0 term that names uveitis / scleritis / episcleritis and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
H30*	Chorioretinal inflammation	56	Moderate	ICD-10-CM code family that names uveitis / scleritis / episcleritis. Chorioretinal inflammation (posterior uveitis).
H15.1*	Episcleritis	12	Moderate	ICD-10-CM code family that names uveitis / scleritis / episcleritis.
H20*	Iridocyclitis	38	High	ICD-10-CM code family that names uveitis / scleritis / episcleritis. Iridocyclitis (anterior uveitis).
H44.11*	Panuveitis	4	High	ICD-10-CM code family that names uveitis / scleritis / episcleritis.
H15.0*	Scleritis	28	Moderate	ICD-10-CM code family that names uveitis / scleritis / episcleritis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Uveitis	608	13,330	3.4	High
Immune-mediated uveitis	174	178	3059.4	High
Iridocyclitis	143	3,086	3.4	High
Vogt-koyanagi-harada disease	90	212	51.9	High
Iritis	56	3,000	1.3	High
Chorioretinitis	38	635	4.5	High

Look-alikes: Infectious uveitis (HSV, CMV, syphilis), endophthalmitis, conjunctivitis, MEK inhibitor retinopathy (combination BRAF/MEK).

Optic neuritis (OPTIC_NEURITIS)

Organ system: Ocular

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Optic nerve disorder	Marked decrease in visual acuity (best corrected visual acuity worse than 20/40 or more than 3 lines of decreased vision from known baseline, up to 20/200)	Moderate	CTCAE v6.0 term that names optic neuritis and grades its severity. Broader term.

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Optic neuritis (new in v6.0)	Severe or medically significant but not immediately sight-threatening; decrease in visual acuity (best corrected visual acuity worse than 20/40 or more than 3 lines of decreased vision from known baseline, up to 20/200) in the affected eye; oral steroids indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	High	CTCAE v6.0 term that names optic neuritis and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
H46*	Optic neuritis	10	High	ICD-10-CM code family that names optic neuritis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Optic neuritis	264	9,000	2.1	High
Optic neuropathy	59	2,071	2.1	High
Immune-mediated optic neuritis	25	26	1758.3	High

Look-alikes: Ischemic optic neuropathy, papilledema, MS or NMOSD, compressive lesion.

Autoimmune hemolytic anemia (AIHA)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Hemolytic anemia (new in v6.0)	>2 units of PRBCs per day indicated	High	CTCAE v6.0 term that names autoimmune hemolytic anemia and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D59.10	Autoimmune hemolytic anemia, unspecified	1	Moderate	ICD-10-CM code family that names autoimmune hemolytic anemia.
D59.12	Cold autoimmune hemolytic anemia	1	Moderate	ICD-10-CM code family that names autoimmune hemolytic anemia.
D59.0	Drug-induced autoimmune hemolytic anemia	1	High	ICD-10-CM code family that names autoimmune hemolytic anemia.
D59.13	Mixed type autoimmune hemolytic anemia	1	Moderate	ICD-10-CM code family that names autoimmune hemolytic anemia. Mixed type AIHA.
D59.19	Other autoimmune hemolytic anemia	1	Low	ICD-10-CM code family that names autoimmune hemolytic anemia.
D59.11	Warm autoimmune hemolytic anemia	1	Moderate	ICD-10-CM code family that names autoimmune hemolytic anemia.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Autoimmune haemolytic anaemia	422	3,366	10.1	High
Haemolytic anaemia	298	8,876	2.4	Moderate
Cold type haemolytic anaemia	18	229	6	High
Warm type haemolytic anaemia	10	94	8.4	High

Look-alikes: Microangiopathic hemolysis (TMA), G6PD deficiency with oxidant drugs, transfusion reaction, marrow infiltration.

Immune thrombocytopenia (ITP)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Thrombocytopenia (new in v6.0)	<50,000 - 10,000/mm3; <50.0 - 10.0 x 10^9/L; transfusion indicated	Moderate	CTCAE v6.0 term that names immune thrombocytopenia and grades its severity (new in v6.0). New in v6.0 as a disorder term; also chemotherapy-related.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D69.3	Immune thrombocytopenic purpura	1	High	ICD-10-CM code family that names immune thrombocytopenia.
D69.59	Other secondary thrombocytopenia	1	Low	ICD-10-CM code family that names immune thrombocytopenia.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Immune thrombocytopenia	570	4,416	10.4	High

Look-alikes: Chemotherapy myelosuppression, marrow infiltration, HIT, DIC, TTP.

Aplastic anemia / marrow failure (APLASTIC_ANEMIA)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Aplastic anemia (new in v6.0)	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated	High	CTCAE v6.0 term that names aplastic anemia / marrow failure and grades its severity (new in v6.0).
Bone marrow hypocellular	Severely hypocellular or >50 - <=75% reduction cellularity from normal for age	Moderate	CTCAE v6.0 term that names aplastic anemia / marrow failure and grades its severity. Marrow failure.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D60*	Acquired pure red cell aplasia [erythroblastopenia]	4	Moderate	ICD-10-CM code family that names aplastic anemia / marrow failure.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
				Acquired pure red cell aplasia.
D61.9	Aplastic anemia, unspecified	1	Moderate	ICD-10-CM code family that names aplastic anemia / marrow failure.
D61.1	Drug-induced aplastic anemia	1	High	ICD-10-CM code family that names aplastic anemia / marrow failure.
D61.811	Other drug-induced pancytopenia	1	Low	ICD-10-CM code family that names aplastic anemia / marrow failure.
D61.89	Other specified aplastic anemias and other bone marrow failure syndromes	1	Moderate	ICD-10-CM code family that names aplastic anemia / marrow failure. Other specified aplastic anemias.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Bone marrow failure	400	20,194	1.4	High
Aplasia pure red cell	188	3,394	4.1	High
Aplastic anaemia	138	5,175	1.9	High
Autoimmune aplastic anaemia	5	13	44	High

Look-alikes: Chemotherapy myelosuppression, marrow infiltration by tumor, HLH.

Thrombotic microangiopathy (TTP / HUS) (TMA)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Hemolytic uremic syndrome	Laboratory findings with clinical consequences (e.g., renal insufficiency, petechiae)	High	CTCAE v6.0 term that names thrombotic microangiopathy and grades its severity.
Thrombotic microangiopathy (new in v6.0)	Transfusion or IV intervention indicated	High	CTCAE v6.0 term that names thrombotic microangiopathy and grades its severity (new in v6.0).
Thrombotic thrombocytopenic purpura	Laboratory findings with clinical consequences (e.g., renal insufficiency, petechiae)	High	CTCAE v6.0 term that names thrombotic microangiopathy and grades its severity.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D59.30	Hemolytic-uremic syndrome, unspecified	1	Moderate	ICD-10-CM code family that names thrombotic microangiopathy.
D59.39	Other hemolytic-uremic syndrome	1	Moderate	ICD-10-CM code family that names thrombotic microangiopathy.
M31.19	Other thrombotic microangiopathy	1	High	ICD-10-CM code family that names thrombotic microangiopathy. Other thrombotic microangiopathy (includes acquired TTP).
M31.10	Thrombotic microangiopathy, unspecified	1	High	ICD-10-CM code family that names thrombotic microangiopathy.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Thrombotic microangiopathy	260	9,560	2	High
Thrombotic thrombocytopenic purpura	97	3,485	2	High
Haemolytic uraemic syndrome	35	2,315	1.1	High
Renal-limited thrombotic microangiopathy	17	223	5.8	High
Atypical haemolytic uraemic syndrome	13	796	1.2	High

Look-alikes: DIC, malignant hypertension, chemotherapy (gemcitabine, mitomycin) or calcineurin inhibitor TMA.

Hemophagocytic lymphohistiocytosis (HLH)

Organ system: Hematologic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Secondary hemophagocytic lymphohistiocytosis (new in v6.0)	Severe or medically significant but not immediately life threatening; hospitalization or prolongation of existing hospitalization indicated	High	CTCAE v6.0 term that names hemophagocytic lymphohistiocytosis and grades its severity (new in v6.0).

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D76.1	Hemophagocytic lymphohistiocytosis	1	High	ICD-10-CM code family that names hemophagocytic lymphohistiocytosis.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Haemophagocytic lymphohistiocytosis	849	7,131	9.5	High
Macrophage activation	2	34	4.4	High

Look-alikes: Infection- or lymphoma-associated HLH, sepsis, CRS.

Sarcoid-like granulomatous reaction (SARCOID_LIKE)

Organ system: Systemic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Immune system disorders - Other, specify	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Moderate	CTCAE v6.0 term that names sarcoid-like granulomatous reaction and grades its severity. No dedicated CTCAE term.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D86*	Sarcoidosis	13	Moderate	ICD-10-CM code family that names sarcoid-like granulomatous reaction. Sarcoidosis; ICI sarcoid-like reaction usually coded here.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Sarcoidosis	209	4,690	3.3	High
Pulmonary sarcoidosis	92	1,091	6.5	High
Cutaneous sarcoidosis	42	418	7.9	High
Sarcoidosis of lymph node	8	53	12.5	High
Cardiac sarcoidosis	7	228	2.2	High
Ocular sarcoidosis	4	119	2.4	High

Look-alikes: Disease progression (hilar nodes mimic metastasis), TB or fungal granulomas.

Immune-related adverse event, organ not specified (IRAE_UNSPECIFIED)

Organ system: Systemic

CTCAE v6.0 term	Grade 3	Specificity	Rationale
Autoimmune disorder	Autoimmune reactions involving major organ (e.g., colitis, anemia, myocarditis, kidney)	Moderate	CTCAE v6.0 term that names immune-related adverse event, organ not specified and grades its severity. Catch-all autoimmune term.
Immune system disorders - Other, specify	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of existing hospitalization indicated; limiting self-care ADL or severe impact on age-appropriate normal daily activity (pediatric)	Low	CTCAE v6.0 term that names immune-related adverse event, organ not specified and grades its severity. Catch-all.

ICD-10-CM	Description	Billable codes	Specificity	Rationale
D89.9	Disorder involving the immune mechanism, unspecified	1	Low	ICD-10-CM code family that names immune-related adverse event, organ not specified.
D89.89	Other specified disorders involving the immune mechanism, not elsewhere classified	1	Low	ICD-10-CM code family that names immune-related adverse event, organ not specified. Other specified disorders involving the immune mechanism.

FAERS reaction term	ICI reports	All reports	PRR	Specificity
Immune-mediated adverse reaction	1,130	1,442	254.7	Moderate
Autoimmune disorder	757	9,445	6.1	Moderate
Immune-mediated endocrinopathy	136	140	2391.3	Moderate

Look-alikes: Use only when the organ is unknown; organ-specific categories take precedence.

6. Drug-cause companion codes

Code	Description	Role	Note
T45.1X5*	Adverse effect of antineoplastic and immunosuppressive drugs	companion	Adverse effect of antineoplastic and immunosuppressive drugs. The code used for ICI adverse effects before 2024-10-01; also covers chemotherapy and immunosuppressants.
T45.AX5*	Adverse effect of immune checkpoint inhibitors and immunostimulant drugs	companion	Adverse effect of immune checkpoint inhibitors and immunostimulant drugs. Added in FY2025 (effective 2024-10-01). Use for encounters on or after that date.
T45.AX1*	Poisoning by immune checkpoint inhibitors and immunostimulant drugs, accidental (unintentional)	lookalike	Poisoning (overdose), not an adverse effect at therapeutic dose; listed so analysts do not confuse it.

An ICD-10-CM condition code alone does not attribute the condition to a checkpoint inhibitor. Attribution in coded data depends on a drug-cause code on the same encounter: T45.AX5- for encounters on or after 1 October 2024, and T45.1X5- before that date. T45.1X5- also covers chemotherapy and immunosuppressants, so analyses before October 2024 cannot separate checkpoint inhibitor toxicity from chemotherapy toxicity by code alone.

7. Limitations

A code is not a diagnosis of an irAE. An ICD-10-CM code such as K52.1 (toxic gastroenteritis and colitis) says what the clinician recorded, not that an immune checkpoint inhibitor caused it. Attribution in records usually depends on a companion drug-cause code (T45.AX5- or T45.1X5-), clinical notes, and timing. irAE-Map supplies the code lists; it does not validate them against chart review. Sensitivity and positive predictive value of these code sets in real EHR data are unknown until someone measures them (see NEXT_STEPS).

The drug-cause code changed on 2024-10-01. T45.AX5- (adverse effect of immune checkpoint inhibitors and immunostimulant drugs) entered ICD-10-CM in FY2025. Before that date, ICI adverse effects were coded with T45.1X5-, which also covers chemotherapy and immunosuppressants. Analyses spanning that date must use both, and T45.1X5- alone cannot separate ICI from chemotherapy toxicity. T45.AX5- also covers "immunostimulant drugs" beyond checkpoint inhibitors.

Many codes are shared with non-immune causes. Codes rated Moderate or Low specificity (for example E03.9 hypothyroidism, unspecified; K52.9 noninfective colitis, unspecified) are common in patients who never received immunotherapy. Their usefulness depends on the analyst restricting to treated patients and a time window.

FAERS counts are reports, not patients or incidence. FAERS is a spontaneous reporting system. Counts reflect reporting behavior, duplicates, stimulated reporting after publications and label changes, and missing data. They cannot estimate how often an irAE happens. ici_prr compares how often a term appears in reports naming an ICI versus all other reports; it is a descriptive over-representation measure, not a validated disproportionality signal (no confidence interval, no stratification, no deduplication).

openFDA cannot tie the reaction to the ICI within a report. The query finds reports in which any listed drug is one of the 13 ICIs (by harmonized generic name or by raw product name, generic or brand). It cannot require that the ICI is the *suspect* drug, because openFDA does

not match drug-level fields within the same drug record. Some counted reports will list the ICI as a concomitant drug.

openFDA coverage of drug names is incomplete. On 2026-10-13, openFDA's harmonized generic-name field returned 0 reports for avelumab, tislelizumab and retifanlimab; we added raw product-name matching (generic and brand names) to recover them. Misspelled or foreign product names are still missed.

FAERS candidate terms come from keyword searches capped at 100 terms each (without an openFDA API key). Rare MedDRA terms outside the top 100 for every keyword can be missed. Re-running code/01_fetch.py with OPENFDA_API_KEY set raises the cap to 1,000. Report counts for the candidate terms themselves are exact (direct per-term queries where the keyword count was not exact).

Category boundaries are a clinical judgment. The categories follow the ASCO 2021 irAE guideline's organ groups, with additions. Other groupings (for example combining thyroid categories, or separating cholangitis from hepatitis) are equally defensible. Some terms legitimately map to more than one category (thyroiditis; the myocarditis-myositis-myasthenia overlap).

MedDRA licensing. CTCAE v6.0 terms and FAERS reaction terms are MedDRA terms. MedDRA is maintained by ICH and licensed through MSSO. irAE-Map republishes term names and the CTCAE LLT codes exactly as they appear in NCI's and FDA's free public releases, and does not redistribute the MedDRA hierarchy or MedDRA PT codes. Users who need MedDRA codes or hierarchy should obtain a MedDRA license.

ICD-10-CM is US-specific and versioned. Codes are valid for FY2027 (2026-10-01 to 2027-09-30). Codes added, deleted or revised in later releases require a new irAE-Map version (v2.0 planned for FY2028). Other countries' ICD-10 modifications differ.

CTCAE grade text is copied verbatim. irAE-Map does not convert grades into code-based rules; ICD-10-CM codes do not carry CTCAE grade except G92.0- (ICANS), which is not an ICI irAE.

Candidates were proposed by rules and drafted with AI assistance. Search rules, code families and draft rationales were produced by Claude (Anthropic) from the public sources. In v1.0 the author accepted the drafted suggestions as a group and set individual rows where needed (data/curation/DECISION_LOG.md); rows without a drafted suggestion are not released. Accepted rows have therefore not each been individually reviewed, and errors in the drafts may remain; please report them through the issue tracker. Rules can still miss relevant codes or terms; the full candidate lists are published so others can check what was considered.

8. Use of AI assistance

Claude (Anthropic) wrote the data-processing code, the candidate search rules, and the draft role, specificity and rationale for candidates. The author made the mapping decisions as described in Section 3.3, verified sample rows against the sources, and is responsible for the content.

9. Versioning and maintenance

irAE-Map uses semantic versioning. ICD-10-CM changes every 1 October; version 2.0, aligned to the FY2028 code files, is planned for October 2027 after CMS and CDC publish them. A Python package, iraemap, that applies the table to a user's codes is planned (version 0.1, March 2027). Corrections and suggestions are welcome through the repository's issue tracker.

10. How to cite

Onyekachi Innocent Ike-Okpe. irAE-Map: codebook and mapping rationale, version 1.0.0. Zenodo. https://doi.org/10.5281/zenodo.23351504. The dataset: https://doi.org/10.5281/zenodo.23351502.